

SENDAI, Japan

WMO: 475900

Lat: 38.2619N

Lon: 140.8967E

Elev: 44

StdP: 100.80

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	-2.6	-10.7	1.5	1.5	-9.6	1.7	1.7	12.4	2.9	11.1	2.0	3.0	340	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	31.7	24.5	30.2	24.1	28.8	23.4	25.5	29.4	25.0	28.5	24.4	27.6	3.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.5	19.5	27.3	23.9	18.9	27.0	23.4	18.2	26.5	78.7	29.6	76.4	28.6	74.1	27.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.4	7.3	-5.2	34.4	1.2	1.2	-6.1	35.3	-6.8	36.0	-7.5	36.7	-8.4	37.6		
			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-6.4	26.7	1.0	0.7	-7.1	27.2	-7.6	27.6	-8.2	28.0	-8.9	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	2.0	2.5	5.5	10.8	15.9	19.4	23.5	24.9	21.4	15.9	9.8	4.6
	DBStd	8.44	2.31	2.57	3.04	3.32	3.00	2.70	3.14	2.85	2.99	2.88	3.16	2.79
	HDD10.0	844	249	210	143	29	1	0	0	0	0	1	43	168
	HDD18.3	2454	507	443	397	226	87	18	2	0	6	86	257	426
	CDD10.0	1967	0	0	4	54	183	282	418	462	343	184	36	1
	CDD18.3	536	0	0	0	1	10	51	161	203	100	11	0	0
	CDH23.3	3258	0	0	0	3	47	171	1049	1509	456	23	0	0
	CDH26.7	874	0	0	0	0	8	26	302	453	83	1	0	0
Wind	WSAvg	3.2	3.5	3.6	3.8	3.6	3.2	2.8	2.5	2.5	2.8	3.0	3.2	3.4
Precipitation	PrecAvg	1243	43	42	72	91	111	139	160	163	191	121	65	38
	PrecMax	1798	148	140	185	217	224	281	445	444	639	347	243	168
	PrecMin	816	2	2	3	4	28	50	27	17	12	8	6	3
	PrecStd	214	33	31	41	48	45	54	90	107	114	80	49	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	13.2	17.9	23.3	27.7	29.5	32.9	34.0	31.1	25.9	19.6	14.8
		MCWB	7.1	7.5	9.9	13.5	16.6	20.9	24.6	25.0	23.8	19.2	13.5	10.5
	2%	DB	8.7	10.4	14.9	20.4	24.4	27.0	31.3	32.2	28.9	23.4	17.9	12.6
		MCWB	4.6	6.0	8.8	12.4	15.9	20.1	24.3	25.0	22.9	18.0	12.8	8.7
	5%	DB	7.1	8.4	12.7	18.4	22.5	25.2	29.8	30.7	27.2	21.8	16.4	10.7
		MCWB	3.5	4.5	7.6	11.6	15.4	19.4	24.0	24.6	22.1	17.0	12.1	7.1
	10%	DB	5.8	6.8	10.8	16.5	20.8	23.7	28.3	29.3	25.9	20.5	15.0	9.2
		MCWB	2.5	3.3	6.4	11.0	14.8	19.0	23.4	24.2	21.9	16.0	11.5	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	8.7	11.6	16.7	19.3	23.4	25.9	26.6	25.1	21.3	15.6	11.5
		MCDB	10.2	11.8	15.1	19.2	23.2	26.6	30.3	31.3	28.0	23.7	17.3	13.7
	2%	WB	5.2	6.5	9.9	14.2	18.1	21.8	25.2	25.8	24.4	19.5	14.4	9.2
		MCDB	7.7	9.6	13.3	17.8	21.7	24.8	29.4	29.9	27.0	21.7	16.4	11.8
	5%	WB	4.0	5.0	8.5	13.1	17.0	20.8	24.6	25.3	23.7	18.1	13.1	7.6
		MCDB	6.5	8.0	11.8	16.5	20.3	23.6	28.5	28.9	26.2	20.5	15.5	10.5
	10%	WB	2.8	3.7	7.0	11.9	16.1	19.8	24.0	24.8	22.8	17.1	11.8	6.1
		MCDB	5.4	6.5	10.5	15.0	19.1	22.5	27.4	28.1	25.2	19.7	14.4	8.8
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.7	7.7	8.2	7.6	6.1	5.7	5.9	6.2	7.2	7.5	6.5
		MCDBR	8.1	9.2	10.7	11.4	10.4	8.7	8.1	8.0	7.6	8.3	9.4	8.7
		MCWBR	5.5	6.2	6.9	6.2	5.2	4.0	3.3	3.2	3.7	4.8	6.1	6.1
	5% WB	MCDBR	8.0	9.0	9.9	9.4	8.4	6.9	7.2	7.0	6.3	6.7	8.4	8.7
		MCWBR	6.0	6.6	7.3	6.2	5.0	3.6	3.3	3.1	3.4	4.7	6.3	6.5
Clear-Sky Solar Irradiance	taub	0.375	0.420	0.494	0.544	0.564	0.562	0.571	0.558	0.487	0.443	0.406	0.382	
	taud	2.090	1.970	1.862	1.819	1.827	1.907	1.947	1.974	2.134	2.158	2.168	2.140	
	Ebn at Noon	766	778	758	749	743	744	733	730	761	752	730	726	
	Edh at Noon	118	152	187	207	209	193	184	175	141	125	109	105	
All-Sky Solar Radiation	RadAvg	1.98	2.75	3.69	4.60	5.04	4.85	4.40	4.40	3.63	2.93	2.13	1.68	
	RadStd	0.13	0.19	0.26	0.45	0.50	0.31	0.67	0.55	0.31	0.29	0.14	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page